



Thermodynamics & Thermochemistry

Grade 11 • Chemistry • Session 2





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Yes

No

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Teacher Introduction

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MUTUAL AGREEMENT

- Hi everyone! Welcome to the AI Live Class!
- This is a recorded interactive Session, but all polls, quizzes, hotspots, etc are live and we will provide personalized feedback to you.
- Your complete participation would be evaluated for the leaderboard.
- There are highly qualified Class Teachers in the class so please be assured that all your doubts will be answered.
- I know some concepts are tricky and hence will revise every concept after its completion.
- I request you to be well behaved and not spam the public chat.
- Try to make the best use of this class and enjoy the session with your classmates :)

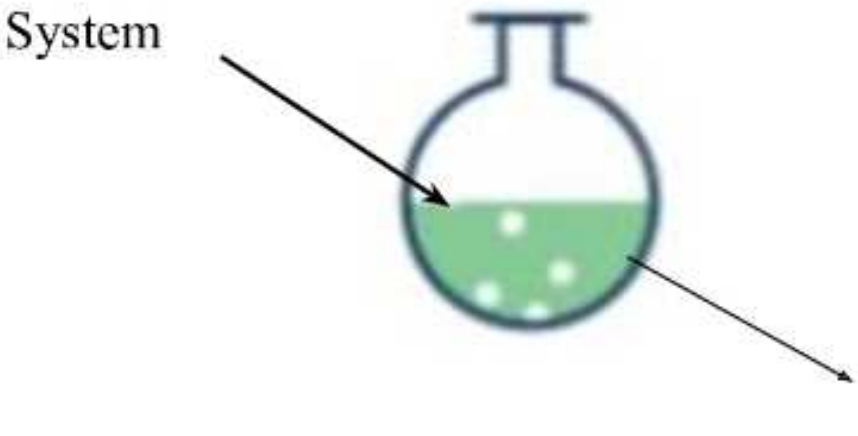


Recap.



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Recap.



Rest is surrounding

Universe = system + surrounding

Open System: Exchanges heat and mass	**Types of System**
Closed System: Exchanges heat but not mass	
Isolated System: Neither heat nor mass exchanged	

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Recap.

Types of thermodynamic variables	State Function
	Path function

Types of thermodynamic properties	Extensive Property
	Intensive Property

Types of processes	Isobaric Process
	Isothermal Process
	Isochoric Process
	Adiabatic Process
	Reversible Process
	Irreversible Process
	Cyclic Process

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Recap.

Zeroth Law of Thermodynamics

If two systems are in thermal equilibrium with third system, then they are also in thermal equilibrium with each other.

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Let's Discuss

What is isothermal process ?



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Learning Outcomes

1. Internal Energy , Heat & Work ✓
2. First Law of Thermodynamics ✓
3. Calculation of Work ✓

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Learning Outcomes

1. Internal Energy , Heat & Work
2. First Law of Thermodynamics
3. Calculation of Work

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Let's Discuss

Can Zeroth law of thermodynamics be derived or it is based on observation ?



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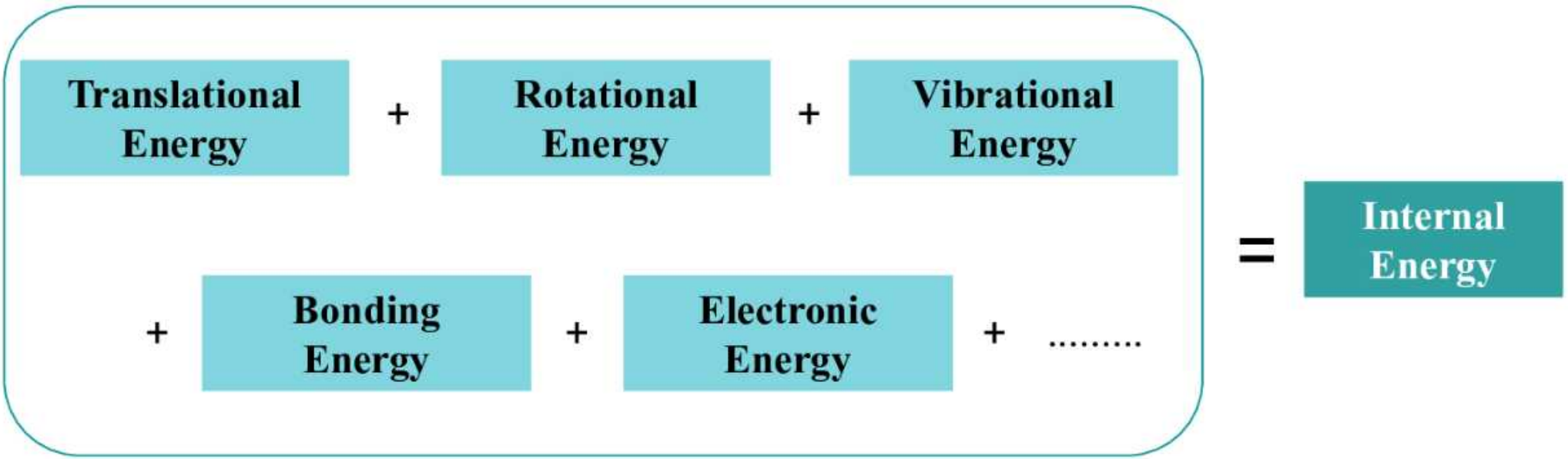
Fundamentals of Thermodynamics

Fundamental quantities of thermodynamics are:

- Internal Energy
- Heat
- Work

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Internal Energy (U or E)



Internal energy is the sum of all microscopic energies possessed by molecules in a system except **gravitational potential energy.**

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Internal Energy (U or E)

Absolute value of internal energy can not be calculated as we generally don't know the values of microscopic energies. **In thermodynamics, we are mainly concerned with the change in U.**

$$\Delta U = U_{final} - U_{initial}$$

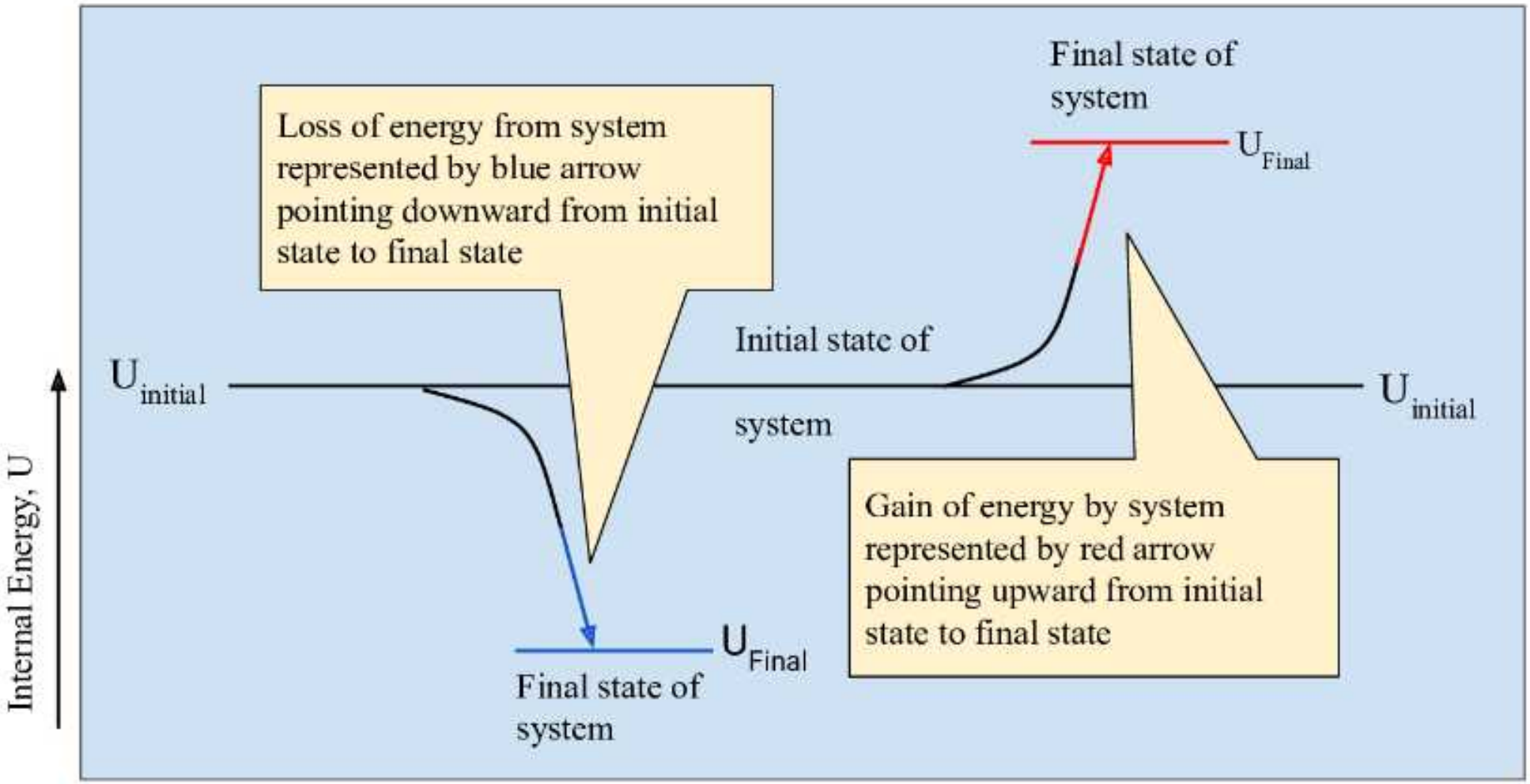
In a chemical reaction, the initial state of the system refers to the reactants and the final state refers to the products.

$\Delta U = U_{final} - U_{initial}$

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Internal Energy (U or E)

Sign Convention for ΔU:



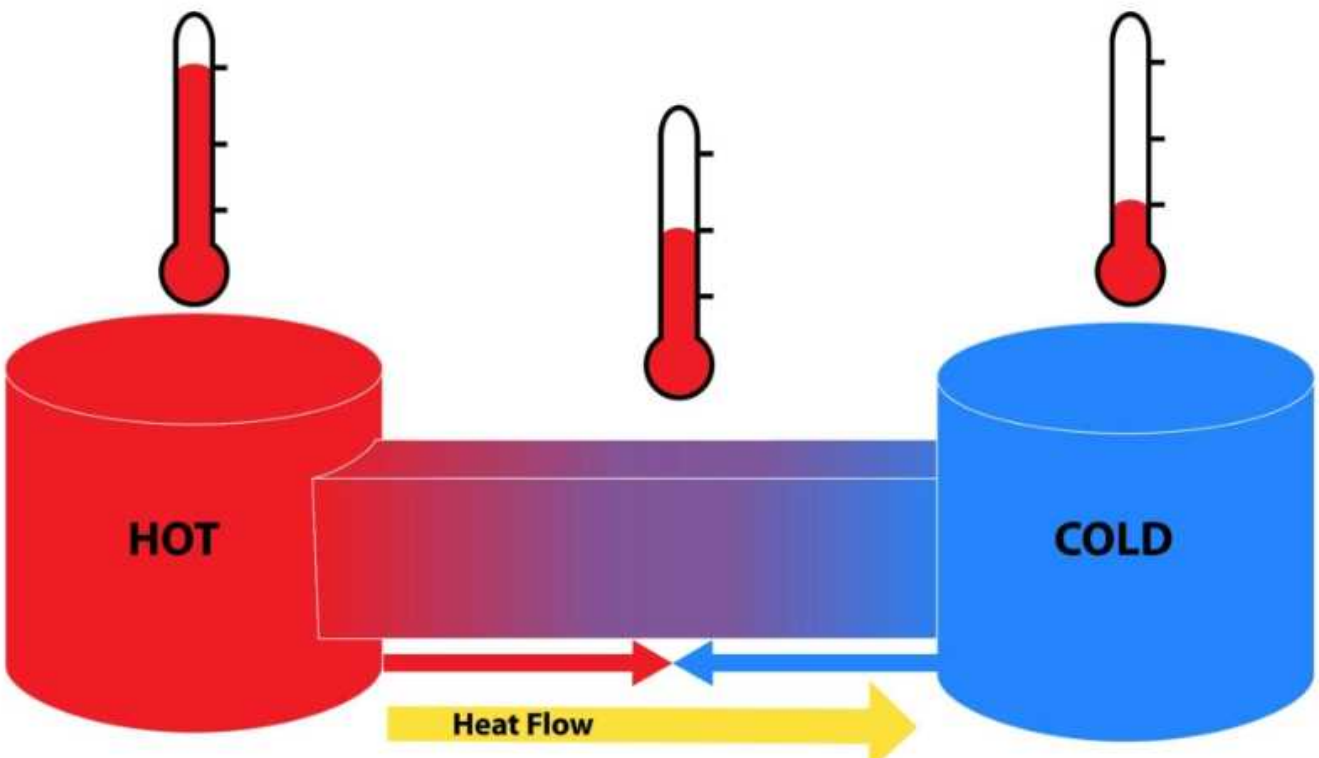
Energy lost to surroundings, internal energy of system decreases, ΔU negative

Energy gained from surroundings, internal energy of system increases, ΔU positive

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Heat(q)

Heat may be defined as the quantity of energy which flows between a system and its surroundings on account of temperature difference. Heat always flows from high temperature to low temperature.



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Heat(q)

Sign Convention:

According to IUPAC system.

(i) Heat flowing into the system q = +ve

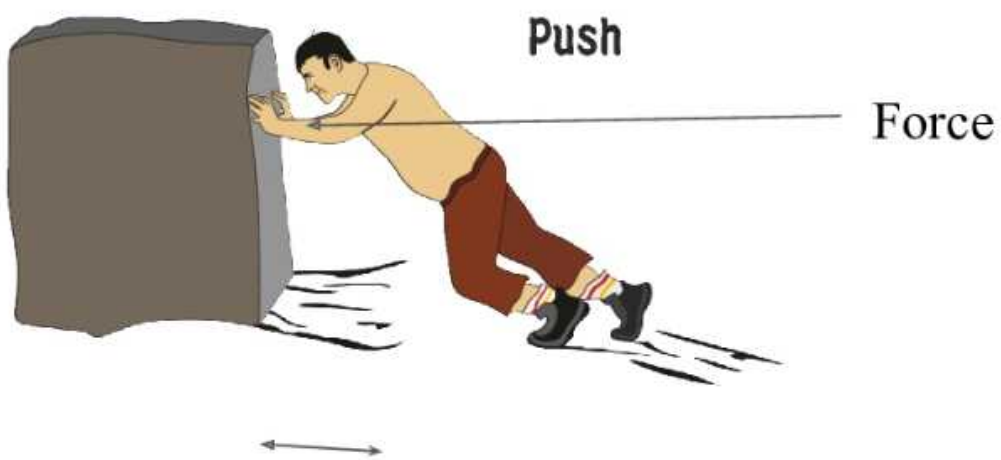
(ii) Heat flowing out of the system, q = -ve

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Work(w)

In thermodynamics, work is generally defined as the force multiplied by the distance. If the displacement of body under the force F is ds, the work done will be,

$$W = F \cdot ds$$



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Work(w)

P-V work

Several things should be noted in the definition of work:

a) Work appears only during a change in state.

b) Work is an algebraic quantity. Sign gives idea about the direction.

c) In SI system of units, work is expressed in Joule or Kilojoule.

(1 J = 1 N m, 1 L-atm = 101.3 J, 1 L-bar = 100 J).

d) Work appears at the boundary of the system.

1 N-m

1 J

l-atm

l-bar

Sign Convention: According to IUPAC system.

(i) Work done **by** the system w = -ve

(ii) Work done **on** the system, w = +ve

1 l-atm = 101.3 J

1 l-bar = 100 J

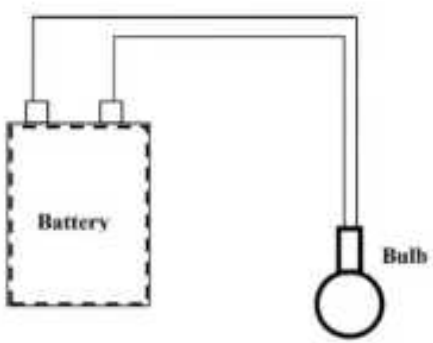
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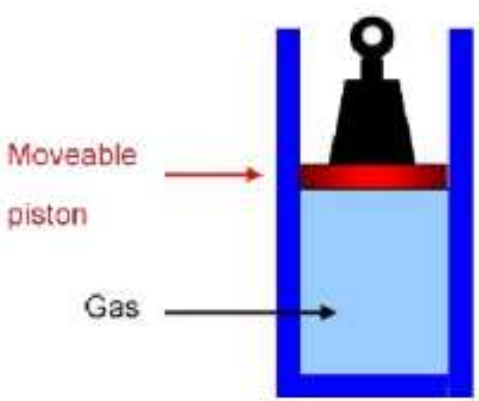
Work(w)

Types of Work

- Gravitational Work
- Electrical Work
- Mechanical Work

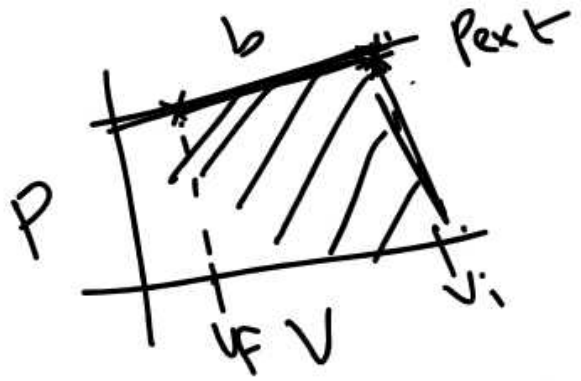






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Work(w)

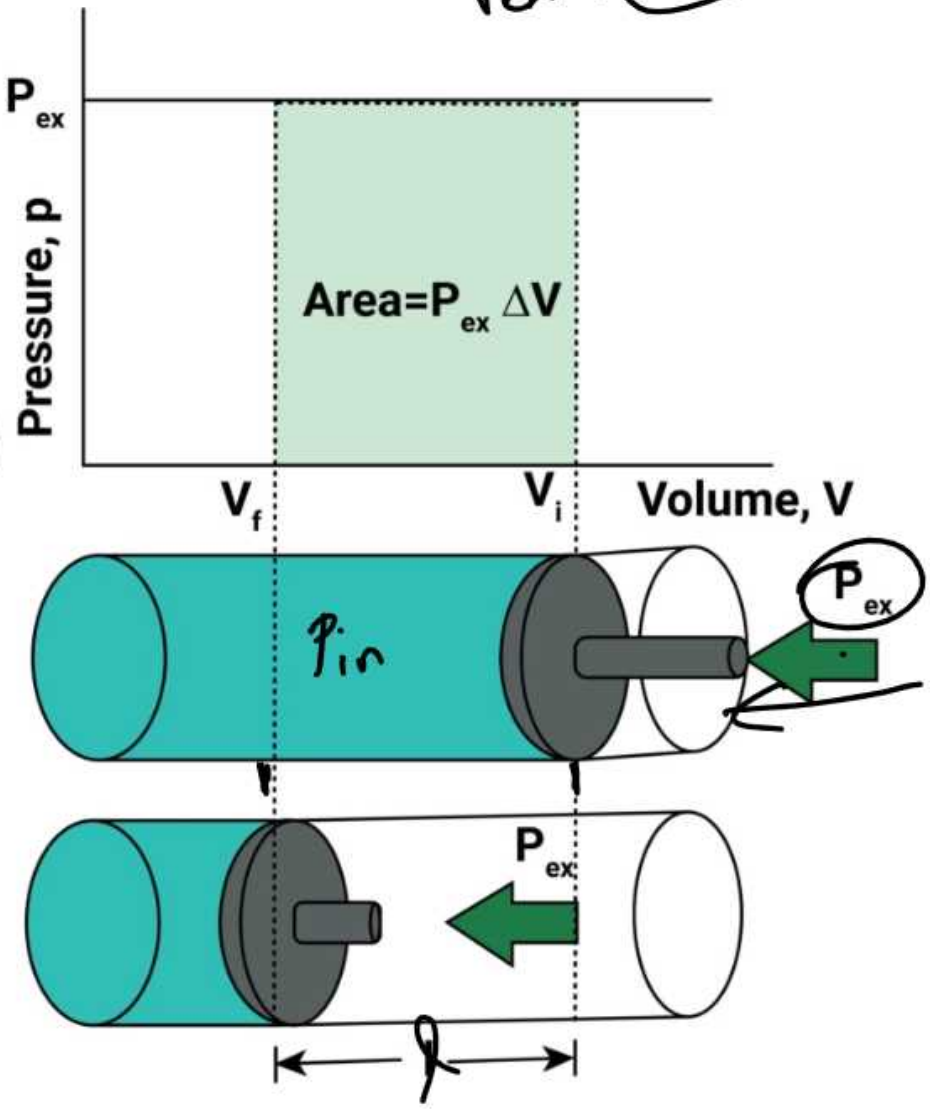


$$W = f \times ds$$

$$P_{ex} \times dV$$

P-V Work

Let us consider a cylinder which contains one mole of an ideal gas fitted with a frictionless piston undergoing process as shown.



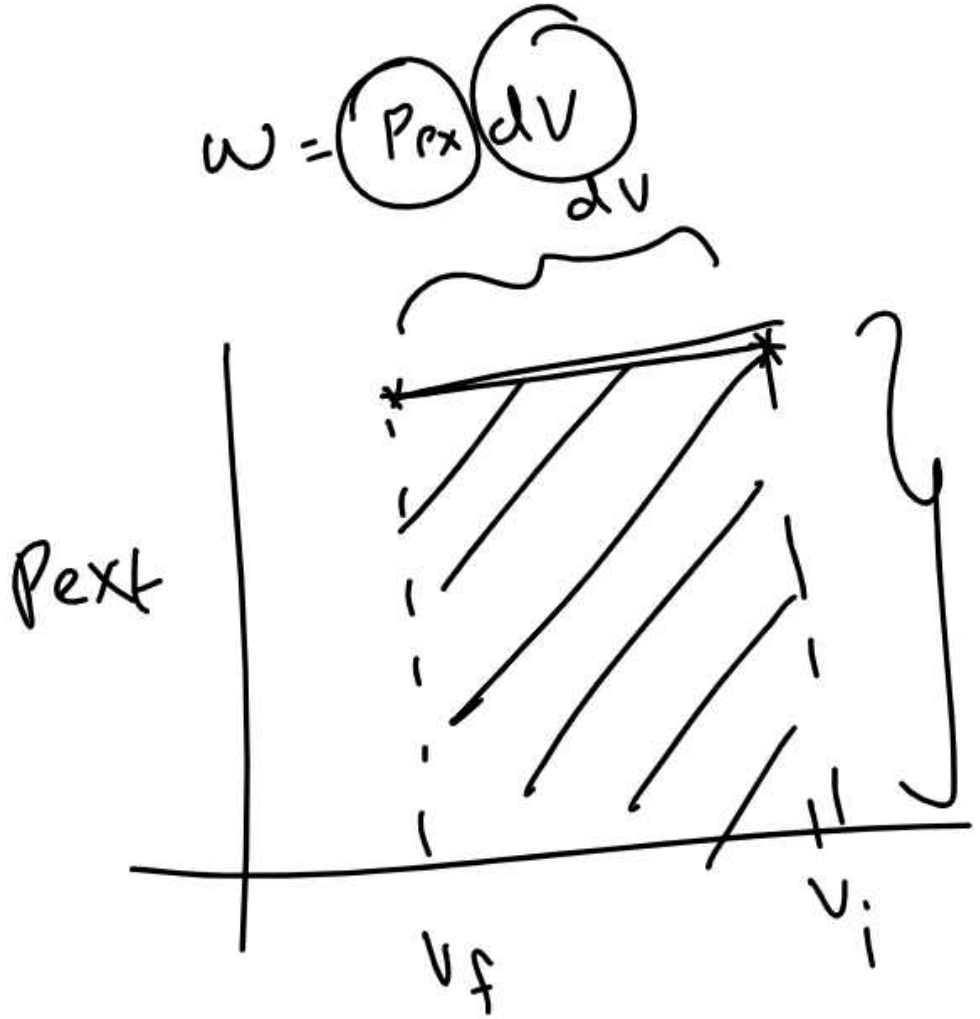
W done on the system.

$P_{ext} > P_{in}$ (Compression)

$W = P_{ex} dV$

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$W = P_{ext} \, dV$



$W = P_{ext} \times dV$

Work(w)

P-V Work

Volume change = l x A = $\Delta V = (V_f - V_i)$

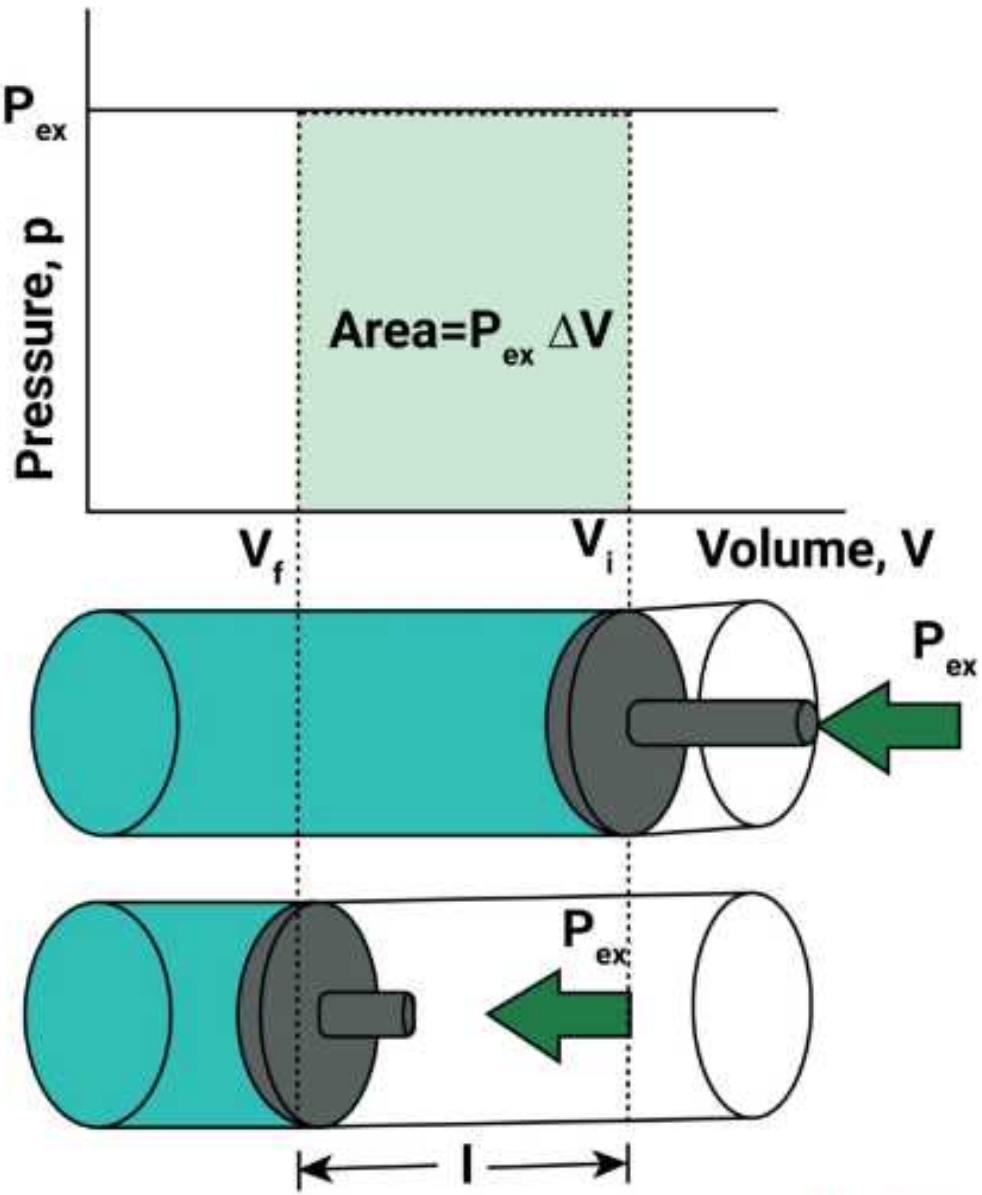
We also know, pressure = $\frac{\text{Force}}{\text{Area}}$

Therefore, force on the poiston = p_{ex} . A

If w is the work done on the system by movement of the piston then

w = force x distance = p_{ex} . A. l

= p_{ex} . $(-\Delta V) = -p_{ex} \Delta V = -p_{ex} (V_f - V_i)$



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Chat Off

Interactivity Pod

Which of the following types of kinetic energies contribute to internal energy?

A Translational

B Translational, Vibrational & Rotational

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Chat On



Interactivity Pod

Which of the following types of kinetic energies contribute to internal energy?

A Translational

B Translational, Vibrational & Rotational

Internal energy consists of the total kinetic energy of all molecules, which means all types, namely translational, rotational & vibrational kinetic energies contribute to internal energy.

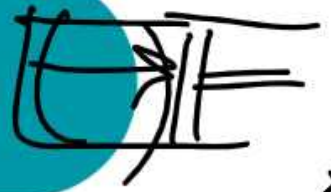
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Let's Solve.



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Let's Solve.



$\Delta V = 0.25 m^3$

$1 J = 1 N-m$

The volume of a gas expands by $0.25\ m^3$ at a constant pressure of $10^3\ N\ m^{-2}$. The work done is equal to

A -2.5 kJ

B -250 J

C -25 J

D -2.5 J

$W = -ve$

$W = +ve$

$P_{ext} = 10^3\ Nm^{-2}$

$\Delta V = 0.25 m^3$

$W = -P_{ext} \Delta V$

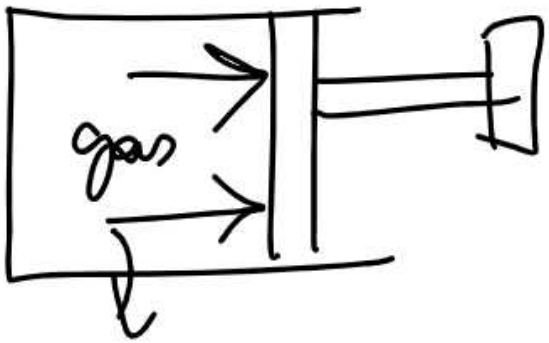
$W =$

$= -10^3 (0.25)$

$= -250\ N-m = -250 J$

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$$W = -P_{ext}\, dV$$



Expansion → work done by the system (gas) (W = -ve)



Contraction → Work done on the system (Compression) (W = +ve)

Let's Solve.

The volume of a gas expands by 0.25 m^3 at a constant pressure of 10^3 N m^{-2}. The work done is equal to

A -2.5 kJ

B -250 J

C -25 J

D -2.5 J

Solution:

Work done, $w = -P_{ex}\Delta V$.

$\therefore W = -10^3 Nm^{-2} x 0.25\ m^3 = -250$ J

So, work done by the gas is -250 J.

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Silent Mode

Chat On

According to IUPAC system, Heat absorbed by the system is +ve.

(T/F)



Discuss among yourselves

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Doubts Time



I will now be taking frequently asked doubts

Doubts



- Can we convert heat into work completely ?

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Quiz.



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Quiz.

Chat Off

An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. The work done in J is

A 1 Joule

B 100 Joule

C 9 Joule

D 900 Joule (−900 J)

$(V_f - V_i)$

$W = -P_{ext}\,\Delta V$

$= -1\text{bar}\,(10-1)$

$= -9$ l-bar

$= -900$ J.

1 l-bar = 1 Joule

10 l-bar = 100 J

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Quiz.

Chat On

An ideal gas is allowed to expand from 1 L to 10 L against a constant external pressure of 1 bar. The work done in J is

A. 1 Joule

B. 100 Joule

C. 9 Joule

D. 900 Joule

Solution

Process is irreversible

$w = -P_{ex}\Delta V = -1.[10 - 1] \times 100\ J = 900\ J$ (1 litre bar = 100 J)

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Quiz.

AIEEE 2004

An ideal gas expands in volume from 1×10^{-3} to $1 \times 10^{-2}\ m^3$ at 300 K against a constant pressure of $1 \times 10^{-5}\ Nm^{-2}$. The work done is

A. 270 kJ

B. − 900 kJ

C. − 900 J

D. 900 kJ

$W = -P_{ext}\Delta V$

$= -10^5 (0.9 \times 10^{-2})$

$= -0.9 \times 10^3$

$= -900\ N\text{-}m = -900\ J$

Quiz.

AIEEE 2004

An ideal gas expands in volume from 1×10^{-3} to 1×10^{-2} m^3 at 300 K against a constant pressure of 1×10^{-5} Nm^{-2}. The work done is

A. 270 kJ

B. − 900 kJ

C. − 900 J

D. 900 kJ

Solution

$$w = -P\Delta V = -10^5[(1 \times 10^{-2}) - (1 \times 10^{-3})]$$
$$= -900 \text{ J}$$

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Milestone Achieved

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Learning Outcomes

1. Internal Energy , Heat & Work
2. First Law of Thermodynamics
3. Calculation of Work

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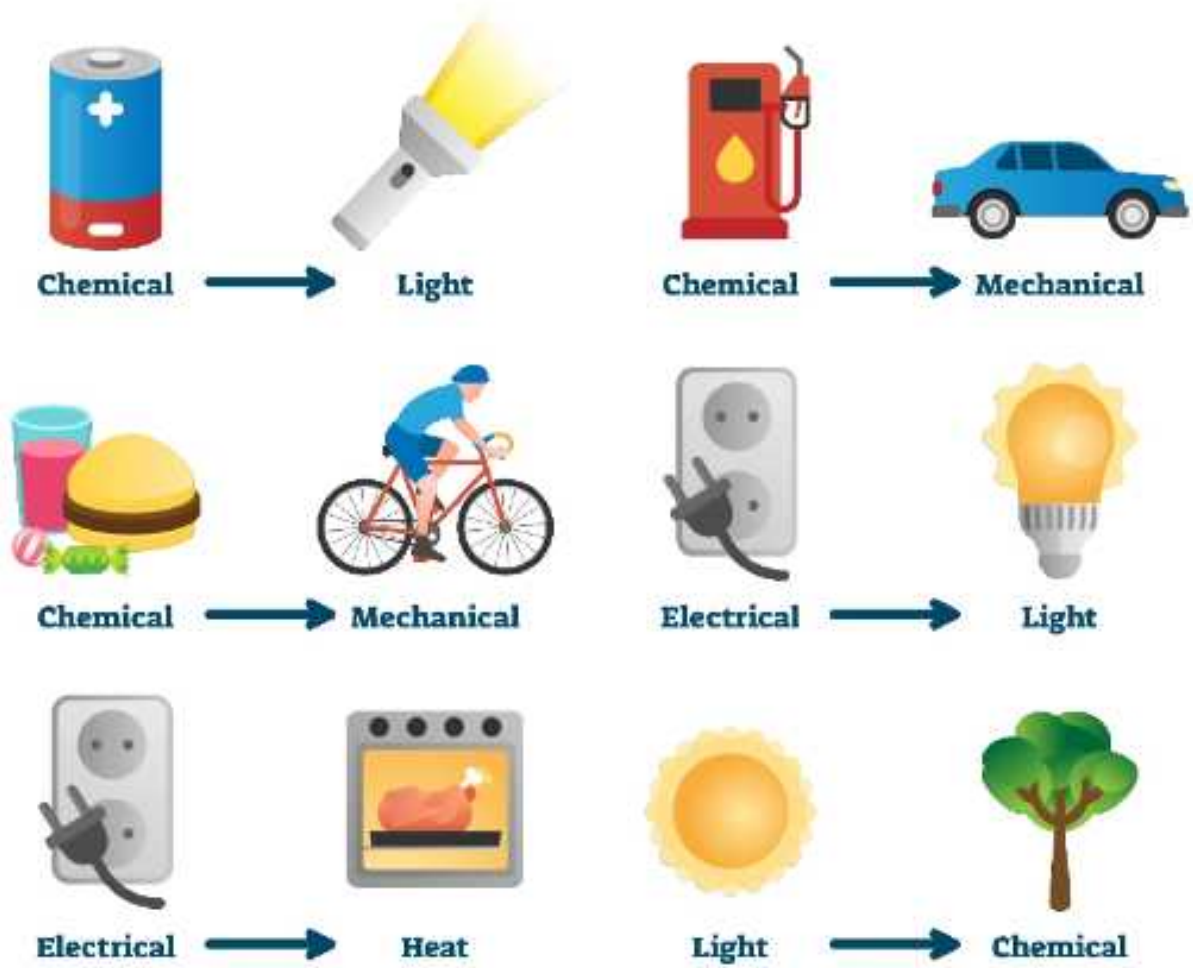
Let's Discuss

What is the IUPAC sign convention for work transfer for a thermodynamic system?

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First Law of Thermodynamics

According to this law **"Energy cannot be created or destroyed but it can be converted from one form to another"**. The energy of an isolated system is constant.



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First Law of Thermodynamics

Mathematical form of first law

$$\Delta U = q + w$$

- ΔU: change in internal energy of the system
- q: heat transfer between system and surroundings
- w: work interaction between system and surroundings

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First Law of Thermodynamics

The following sign conventions are to be followed:

Quantity	Positive Sign (+)	Negative Sign (–)
q	means system **gains** heat	means system **loses** heat
w	means work done **on** system	means work done **by** system
ΔU	means net **gain** of energy by system	means net **loss** of energy by system

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Isochoric Process

$\Delta V = 0$

$w = -P_{ext}\,\Delta V = 0$

$\Delta U = q_v$

Adiabatic Process

$q = 0$

$\Delta U = w$

Isothermal process

$\Delta T = 0$ & $\Delta K.E = 0$

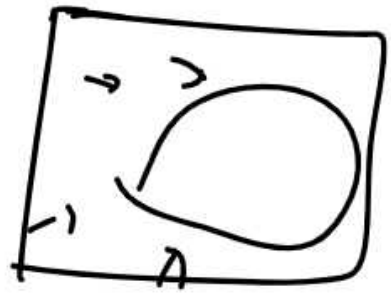
$\Delta U = 0$

Isobaric Process

$\Delta P = 0$

$\Delta U = q + w$

Cyclic process



$\Delta U = 0$

$q \neq 0$ $\quad w \neq 0$

Adiabatic free expansion :-

expansion in vaccum

a) $\Delta U = 0$ $\quad$ b) $\Delta U = w$

$q = 0$

$w = -P_{ext} \Delta V = 0$

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Interactivity Pod

First law of thermodynamics is based on?

A Conservation of energy

B Conservation of mass

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Chat On



Interactivity Pod

First law of thermodynamics is based on?

A Conservation of energy

B Conservation of mass

The first law of thermodynamics is based on the conservation of energy. It deals with work done and heat energy supplied or removed from a system. It basically says that energy supplied to a system is conserved.

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Let's Solve.



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Let's Solve.

A system absorbs 10 kJ of heat and does 4 kJ of work. The internal energy of the system

A. Decreases by 6 kJ

B. Increases by 6 kJ

C. Decreases by 14 kJ

D. Increases by 14 kJ

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Let's Solve.

A system absorbs 10 kJ of heat and does 4 kJ of work. The internal energy of the system

A. Decreases by 6 kJ

B. Increases by 6 kJ

C. Decreases by 14 kJ

D. Increases by 14 kJ

Solution:

As per 1st law,

$\Delta U = q + w$

$q = +10$ kJ (heat is absorbed by the system)

$w = -4$ kJ (work is done by the system)

$\therefore \quad \Delta U = q + w = 10 - 4 \text{ kJ} = +6 \text{ kJ}$

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Silent Mode

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According to First Law of Thermodynamics, $\Delta U = q - w$. (T/F)



Discuss among yourselves

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Doubts Time



I will now be taking frequently asked doubts

Doubts



- When I do work on a piston, it is negative or positive for me ?

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Quiz.



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Quiz.

JEE Mains 2019

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During compression of a spring the work done is 10 kJ and 2 kJ escaped to the surroundings as heat. The change in internal energy, ΔU (in kJ) is:

A. –12

B. –8

C. 8

D. 12

$W = +10\,kJ$

$\Delta U = q + w$

$= -2\,kJ + 10\,kJ$

$= 8\,kJ.$

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Quiz.

JEE Mains 2019

Chat On

During compression of a spring the work done is 10 kJ and 2 kJ escaped to the surroundings as heat. The change in internal energy, ΔU (in kJ) is:

A. -12

B. -8

C. 8

D. 12

Solution

Since the process is compression, work is done on the system, i.e., $w = +10$ kJ

Since the heat is given out of the system, $q = -2$ kJ

Now, $\Delta U = q + w = -2 + 10 = 8$ kJ

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Quiz.

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$q = -45\ J$

A gas has constant pressure in a system. There is a loss of 45 J of heat to the surroundings. 450 J of work is done on to the system. Find the change in internal energy during the process. $W = 450\ J$

A. +405 J

B. −405 J

C. +495 J

D. −495 J

$$\Delta U = -45 + 450$$

$$= +405\ J.$$

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Quiz.

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A gas has constant pressure in a system. There is a loss of 45 J of heat to the surroundings. 450 J of work is done on to the system. Find the change in internal energy during the process.

A. + 405 J

B. – 405 J

C. + 495 J

D. – 495 J

Solution

For the process,

$\Delta U = q + w$

Since heat is lost to the surroundings, $q = -45$ J

Since work is done on the system, $w = +450$ J

Thus, $\Delta U = -45 + 450 = 405$ J

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9. What is the change in internal energy when a gas contracts from 377 mL to 177 mL under a constant pressure of 1520 torr, while at the same time being cooled by removing 124 J heat?

(a) 40.52 J
(b) –83.48 J
(c) –248 J
(d) None of these

1 l-atm ≈ 100 J

1 atm = 760 torr

2 atm.

1520 / 760

0.377 l.

0.177 l.

$\Delta U = q + w$

$= -124\,J + 40\,J$

$w = -P_{ext}\,\Delta V$

$= -2\,atm\,(0.177 - 0.377)$

$= -2\,atm\,(-0.2)$

$= 0.4\,l\text{-}atm$

$= 40\,J$

-124
40

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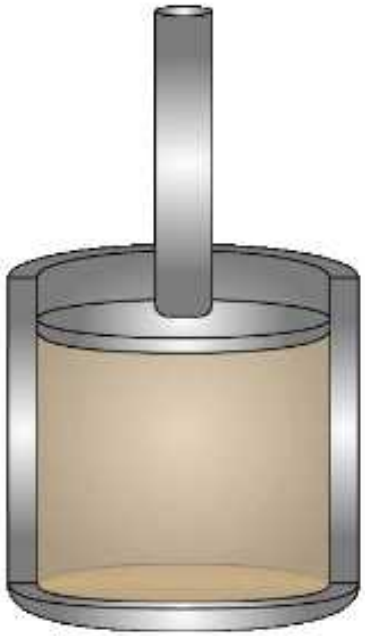


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Learning Outcomes

1. Internal Energy , Heat & Work
2. First Law of Thermodynamics
3. Calculation of Work

Calculation of Work (w)



Air

Compression

$\triangle V$ = –ve

W = +ve

Work is done on the system



Air

Expansion

$\triangle V$ = +ve

W = –ve

Work done by the system

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Calculation of Work (w)

$$W_{rev} = -\int_{V_i}^{V_f} p_{ex}\, dV = -\int_{V_i}^{V_f} (p_{in} \pm dp)\, dV$$

Since $dp \times dV$ is very small we can write

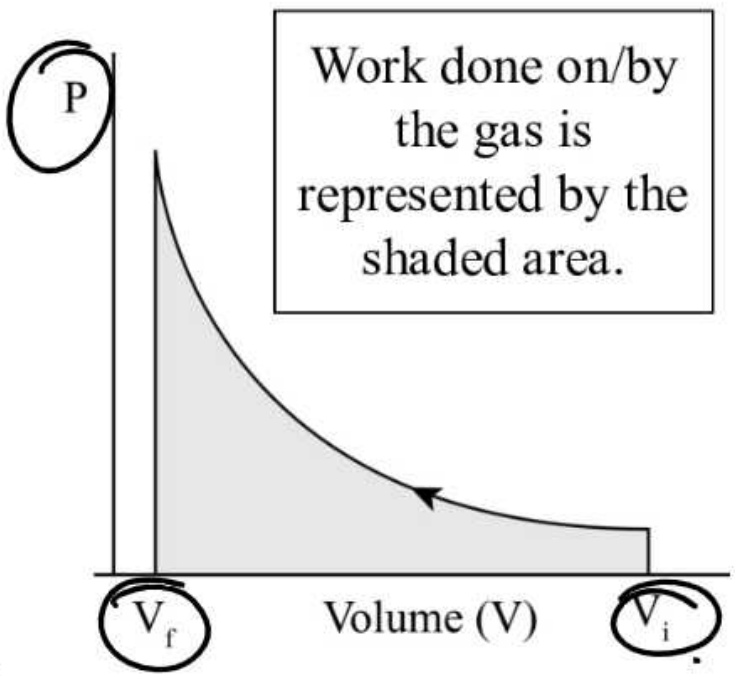
$$w_{rev} = -\int_{V_i}^{V_f} p_{in}\, dV$$

For 'n' mol of an ideal gas, $pV = nRT$

$$\Rightarrow p = \frac{nRT}{V}$$

Therefore, work done can also be calculated as,

$$w_{rev} = -\int_{V_i}^{V_f} nRT\, \frac{dV}{V}$$



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(Isothermal reversible process)

$$w = -\int_{V_i}^{V_f} P_{ext}\, dV$$

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

$$\int \frac{1}{x} dx = \ln x$$

$$= -\int_{V_i}^{V_f} \frac{nRT}{V} dV$$

$$w_{rev} = -2.303 nRT \log \frac{P_i}{P_f}$$

$$= -nRT \int_{V_i}^{V_f} \frac{1}{V} dV = -nRT \ln V \Big]_{V_i}^{V_f}$$

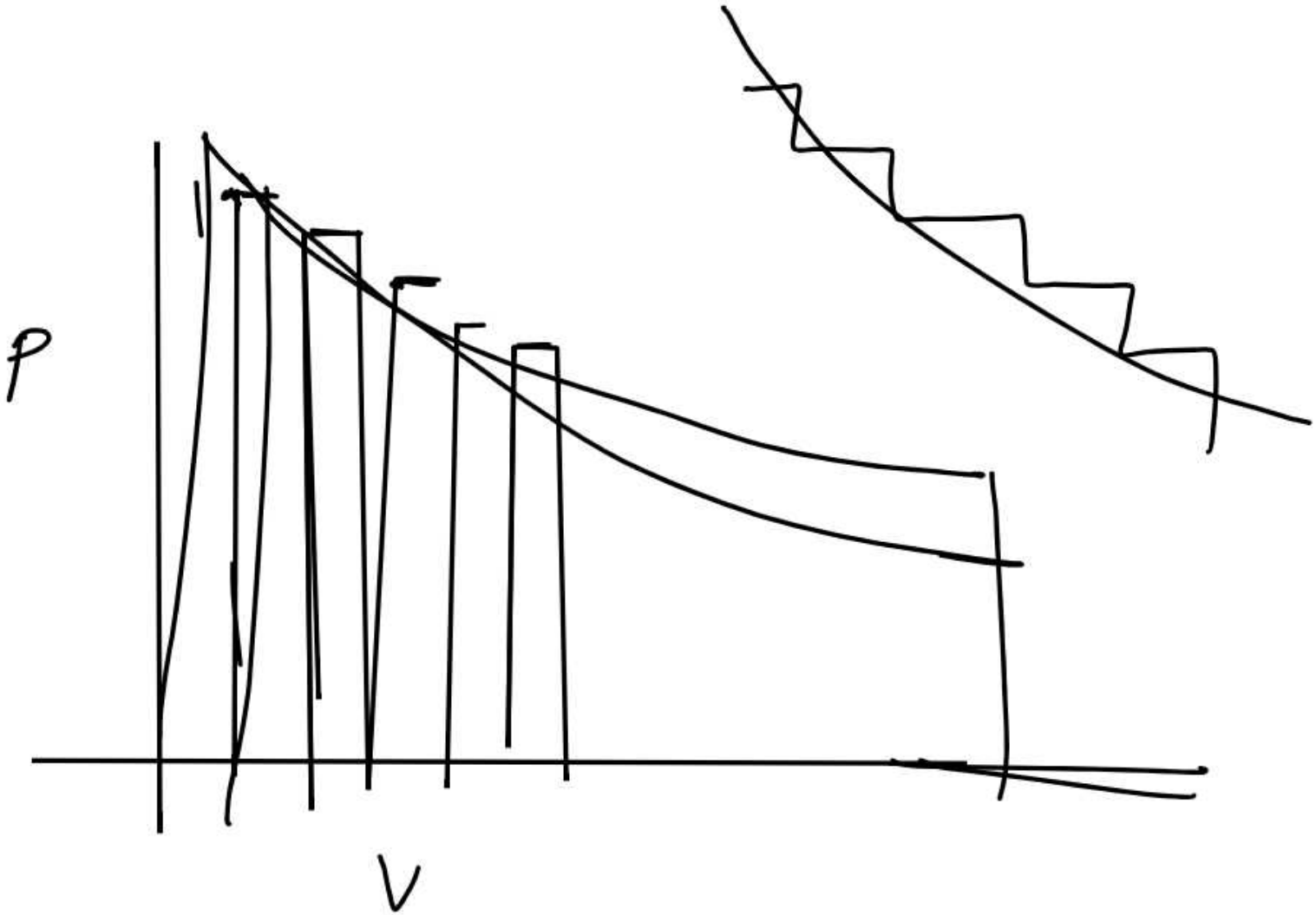
(or)

$$w_{rev} = -2.303 nRT \log \frac{V_f}{V_i}$$

$$= -nRT \ln V_f - \ln V_i$$

$$w_{rev} = -nRT \ln \frac{V_f}{V_i}$$


P
V


P
V

Calculation of Work (w)

Work in Isothermal process

Work in reversible process

$$W_{PV} = -\int_{V_1}^{V_2} P_{ext}\,dV$$

$$P_{gas} = P_{ext} + dP$$

$$P_{gas} \cong P_{ext}$$

$$W_{rev} = -\int_{V_1}^{V_2} P_{gas}\,dV$$

$$W_{rev} = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

Work in irreversible process

$$W_{PV} = -\int_{V_1}^{V_2} P_{ext}\,dV$$

$$P_{ext} = \text{constant}$$

$$W_{irr} = -P_{ext}(V_2 - V_1)$$

Calculation of Work (w)

$\Delta U = q + w$

$\Delta V = 0$



Work in Isochoric process

V = constant

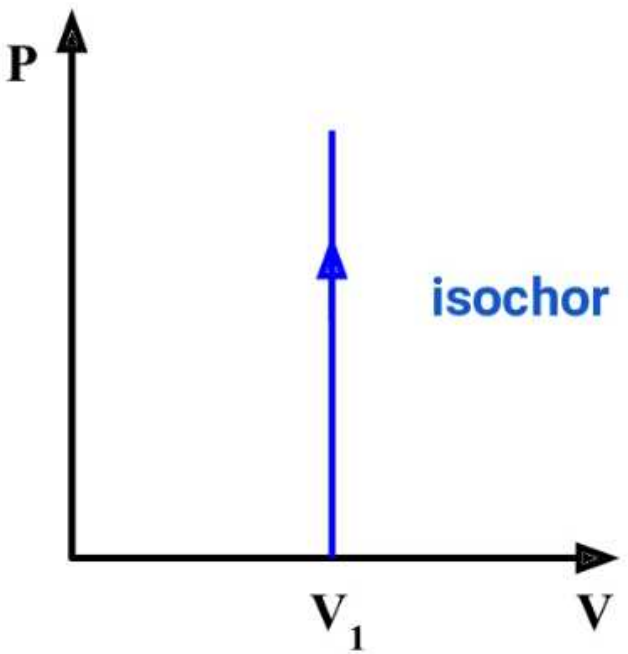
dV = 0

W = 0

From first law of thermodynamics

$\Delta U = q_v$

Heat exchanged at constant volume



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Calculation of Work (w)

Work in Isobaric process

P_{gas} = constant

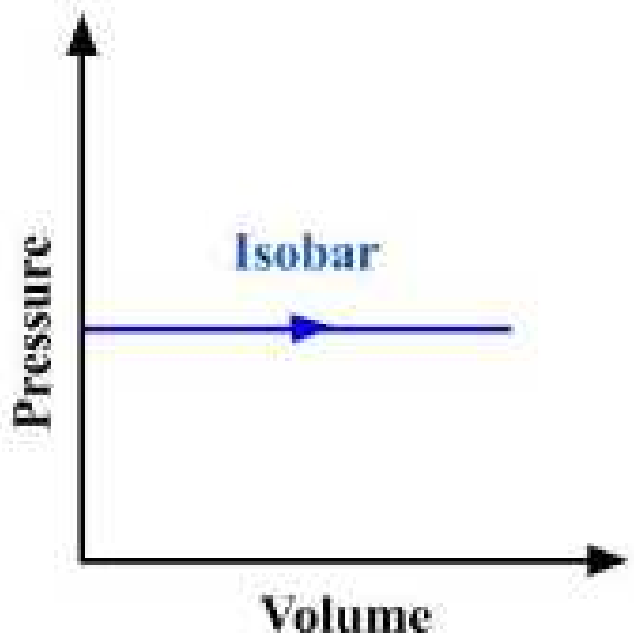
Since pressure of gas is defined during whole of the process and hence the process must be reversible

So $dw = -P_{ext} dV$

$= -P_{gas} dV$

$= -P_{gas} \int_{V_i}^{V_f} dV$

$W = -P(V_f - V_i)$



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Interactivity Pod

For which thermodynamic process work done is zero?

A Isobaric process

B Isochoric process

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Interactivity Pod

For which thermodynamic process work done is zero?

A Isobaric process

B Isochoric process

For isochoric process,

V = constant

$dV = 0$

$W = -PdV = 0$

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Let's Solve.



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Let's Solve.

What is the work done by 1.00 mol of an ideal gas expanding reversibly from a volume of 22.4 L to a volume of 44.8 L at a constant temperature of 273 K?

$$W = -2.303\, nRT \log \frac{V_f}{V_i}$$

$$= -2.303 \times 1\,\text{mol} \times 8.314\,\text{J K}^{-1}\text{mol}^{-1} \times 273\,\text{K} \log \frac{44.8\,\text{L}}{22.4\,\text{L}}$$

$$= -2.303 \times 8.3 \times 273 \log 2$$

$$= -2.303 \times 8.3 \times 273 \times 0.301$$

$$= -1570\,\text{J}$$

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Let's Solve.

What is the work done by 1.00 mol of an ideal gas expanding reversibly from a volume of 22.4 L to a volume of 44.8 L at a constant temperature of 273 K?

Solution:

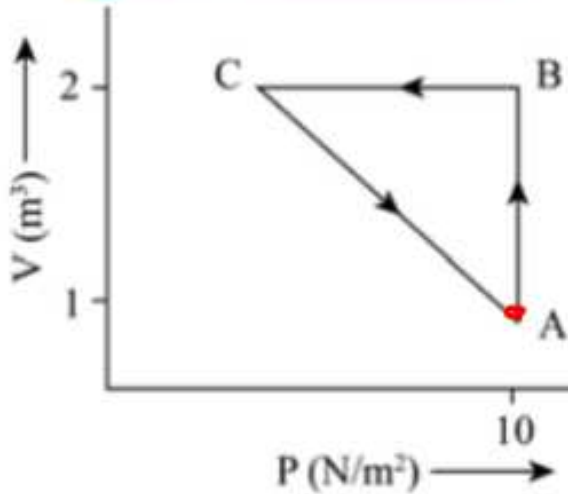
$$w = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

$$w = -(1.00 \cancel{\text{mol}})\left(8.314 \frac{\text{J}}{\cancel{\text{mol}}\,\cancel{\text{K}}}\right)(273 \cancel{\text{K}}) \ln\left(\frac{44.8\ \text{L}}{22.4\ \text{L}}\right)$$

$$= -1570\ \text{J} = -1.57\ \text{kJ}$$

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12. An ideal gas is taken through the cycle A → B → C → A as shown in fig. If the net heat supplied to the gas in cycle is 5 J, the work done by the gas in the process C → A.



(a) –5 J (b) –10 J
(c) –15 J (d) –20 J

13. In a system where ΔE = –51.0 kJ, a piston expanded against a p_{ext} of 1.2 atm giving a change in volume of 32.0 L. What was the change in heat of this system?

(a) –36 kJ (b) –13 kJ
(c) –47 kJ (d) 24 kJ

First Law of Thermodynamics

Doubts Time



I will now be taking frequently asked doubts

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Doubts



- Why work done in isothermal reversible process and isothermal irreversible process is not same ?

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Quiz.



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Quiz.

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What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

A. 10 J

B. 20 J

C. 30 J

D. 40 J

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Quiz.

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What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

A 10 J

B 20 J

C 30 J

D 40 J

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Quiz.

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What is the work done when a gas contracts irreversibly from 377 mL to 177 mL under a constant pressure of 1520 torr while at the same time being cooled by removing 124 J of heat? (Take 1 atm = 760 torr and 1 atm L = 100 J)

Solution

The given process is irreversible. Heat is removed from the system, so its value will be negative, $q = -124$ J.

$$P_{ext} = 1520 \text{ torr} = \frac{1520}{760} \text{atm} = 2 \text{ atm}$$

$$W = -P_{ext}(V_2 - V_1) = -2(0.177 - 0.377)$$

$$W = 0.4 \text{ atm L}$$

$$W = 0.4 \times 100 \text{ J} = 40 \text{ J}$$

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Quiz.

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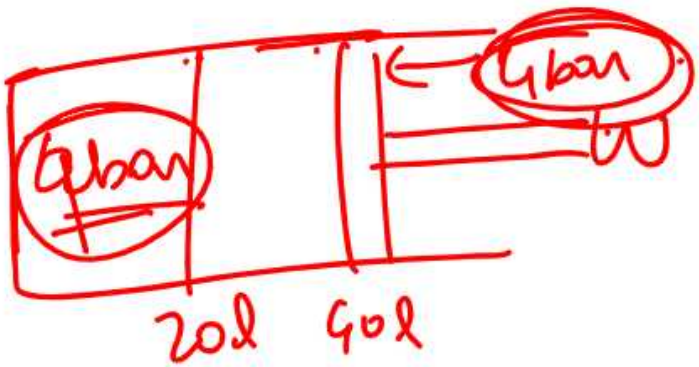
A real gas is subjected to an adiabatic process from (2 bar, 40 L, 300 K) to (4 bar, 30 L, 300 K) against a constant pressure of 4 bar. Calculate the work done for the process. (Take 1 bar L = 100 J)

A. 4000 J

B. 3432 J

C. 40 J

D. 6321 J



$= -P_{ext}\,\Delta V$

$= -4\text{bar}\,(30-40)$

$= +40\text{ l-bar}$

$= 4000\text{ J}$



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Quiz.

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A real gas is subjected to an adiabatic process from (2 bar, 40 L, 300 K) to (4 bar, 30 L, 300 K) against a constant pressure of 4 bar. Calculate the work done for the process. (Take 1 bar L = 100 J)

A. 4000 J

B. 3432 J

C. 40 J

D. 6321 J

Solution

Work done is given by:

$W = -P_{ext}(V_2 - V_1)$

$W = -4(30 - 40)$

$W = 40 \text{ bar L} = 40 \times 100 = 4000 \text{ J}$

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Milestone Achieved

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Summary.



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Summary

First Law of Thermodynamics

Energy can neither be created nor be destroyed.

$\Delta U = q + w$

Quantity	Positive Sign (+)	Negative Sign (–)
For q	System gains heat	System loses heat
w	Work done on system	Work done by system
ΔU	Net gain of energy by system	Net loss of energy by system

Summary

Work in Isothermal reversible process

$$W_{rev} = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

Work in Isothermal irreversible process

$$W_{irr} = -P_{ext}(V_2 - V_1)$$

Work in Isochoric process

$$W = 0$$

Work in Isobaric process

$$W = -P(V_f - V_i)$$

HOMEWORK

(Pick and Solve Based on Session Coverage)

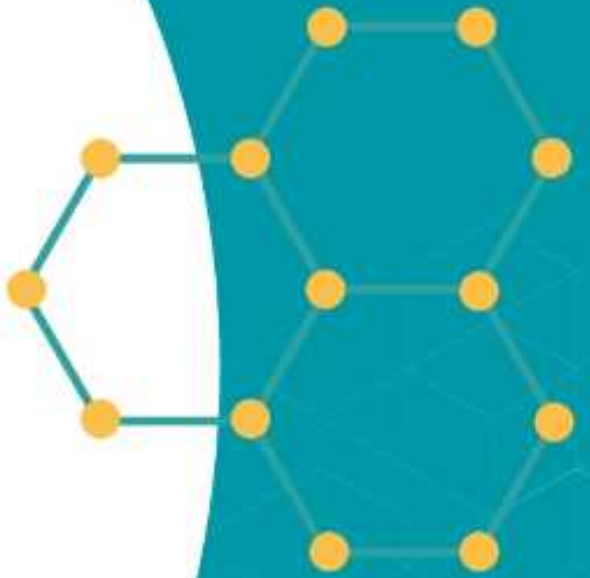
Topic name as per Curriculum tree	Assignment Detail	Tatva Exercise 1 question number
Introduction to thermodynamics	Assignment 1 (question 1, 3, 4, 12)	1, 2, 3, 4, 5, 6,
Types of processes	Assignment 1 (question 5, 6, 7, 8) Assignment 4 (question 13, 16, 17)	
Internal energy	Assignment 1 (question 9, 10, 13, 16) Assignment 2 (question 3, 9, 12)	7, 8, 9
Concept of Heat and work	Assignment 1 (question 2, 11, 14, 15) Assignment 2 (question 10)	10, 11, 12, 13
Zeroth law of thermodynamics	Assignment 2 (question 1)	
First law of thermodynamics	Assignment 2 (question 2, 7, 11, 13, 14)	14, 15, 16, 17
Reversible and irreversible work	Assignment 2 (question 4, 5, 6, 8, 15) Assignment 4 (question 18, 19)	18, 19, 20, 21, 22, 23, 24, 25, 26, 27
Heat Capacity of A System	Assignment 3 (question 1, 2, 3, 4, 12, 13) Assignment 4 (question 14)	28, 29
Adiabatic process and poisson's ratio	Assignment 3 (question 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 17)	30, 31, 32, 33, 34
Enthalpy of A System	Assignment 4 (question 6, 7, 8, 11, 12)	35, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48,
Gibb's Free Energy Change in Reaction	Assignment 6 (question 7, 14)	

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Feedback



Thanks.



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